

1 AML Simulation and portfolio optimization

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1.1 Modeling and simulating Underlying asset dynamics:

1.1.1 Bates AML for SVj1

$$dS_t = \left[\mu_0 + \left(\mu_1 - \frac{1}{2} \right) V_t - (\lambda_0 + \lambda_1 V_t) \bar{k} \right] dt + \sqrt{V_t} \left(\rho dW_{1t} + \sqrt{1 - \rho^2} dW_{2t} \right) + \gamma_s dN_t, \quad (1)$$

$$dV_t = (\alpha - \beta V_t) dt + \sigma \sqrt{V_t} dW_{1t}, \quad (2)$$

(W_{1t}, W_{2t}) independent Brownian motions ($dW_{1t}, dW_{2t} = 0$),

inst. corr. between return and variance shocks is ρ via the dW_{1t} term,

N_t is a Poisson counter with intensity $\lambda_0 + \lambda_1 V_t$,

$\gamma_s \sim \mathcal{N}(\gamma, \delta^2)$ (log-jump size), $\bar{k} \equiv e^{\gamma + \frac{1}{2}\delta^2} - 1$.

$$\arg \max_{\theta} T(\theta) \quad (3)$$

$$T(\theta) = -\log \text{likelihood} \quad (4)$$

$$T(\theta) = - \sum_{i=1}^N p(y_{t+1} y_t | \theta) \quad (5)$$

$$(6)$$

Algorithm 1 Parameter Optimization Loop

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1:  $\theta_{\text{init}} \leftarrow (\mu_0, \mu_1, \rho, \alpha, \beta, \sigma, \lambda_0, \lambda_1, k, dt)$ 
2:  $T_{\text{threshold}} \leftarrow 0.1$ 
3: Initialize inputs
4:  $\{N: \text{Evaluation Budget count, } h: \text{gradient descent step size, } \text{prior}_{\theta} \leftarrow \text{initial prior params}\}$ 
5:  $g \leftarrow 0$ : Intialize gradient estimate
6:  $P = P_{\text{init}} = [p^{(1)}, \dots, p^{(\ell)}]$ : Choose structured direction matrix
7:  $\theta_{\text{old}} = \theta_{\text{init}}$ 
8:  $= \theta_{\text{init}}$ 
9: for  $i = 1$  to  $\ell$  do
10:    $\theta_{\text{new}} \leftarrow \theta_{\text{init}} + hp^{(i)}$ 
11: end for
12:  $i = 0$ 
13: while  $T(\theta) > T_{\text{threshold}} \cap i < N$  do
14:   Update prior $_{\theta_{\text{init}}}$ 
15:    $B = \{s_1..s_{\text{batch\_size}}\}$  (Sample start Idxs of y)
16:    $s_{\text{len}} : \text{Sample Length}$ 
17:    $Y = \{ind \in B | \{y_{ind:(ind+s_{\text{len}})}\}\}$ 
18:   Compute perturbed log-likelihood:
19:   for  $\theta \in \{\theta_{\text{old}}, \theta_{\text{new}}\}$  do
20:      $\{ind \in B | \text{LogLikelihood}_{ind, total} = 0\}$ 
21:     for  $\{y_0, \dots, y_T\} \in \{\{y_{ind:(y_{ind}+s_{\text{len}})}\}\}$  do
22:       for  $y_t \in \{y_0, \dots, y_T\}$  do
23:         Store:  $y_{t-1}$ 
24:          $cf : F(\theta, \phi, \psi = 0) = e^{C(\phi, 0)} G_{\text{prior}_{\theta}}[D(\phi, 0)]$ 
25:          $p(y_{t+1} | Y_t) \approx \frac{\Delta\Phi}{2\pi} \sum_{k=-K}^K cf e^{-i\Phi_k y_{t+1}}, \quad \Phi_k = k \Delta\Phi.$ 
26:          $\text{LogLikelihood}_{total} += \log(p(y_{t+1} | Y_t))$ 
27:       end for
28:        $LL_B \leftarrow \text{LogLikelihood}_{total}$ 
29:     end for
30:      $LL_{\text{Cache}} \leftarrow LL_B$ 
31:   end for
32:    $LL_{\text{old}} \in LL_{\text{Cache}}$ 
33:    $LL_{\text{new}} \in LL_{\text{Cache}}$ 
34:   Update Gradients and Parameters:
35:    $P = [p^{(1)}, \dots, p^{(\ell)}]$ 
36:   for  $i = 1$  to  $\ell$  do
37:      $g \leftarrow g + \frac{(LL_{\text{new}} - LL_{\text{old}})}{h} p^{(i)}$ 
38:   end for
39:    $g \leftarrow \frac{d}{\ell} g$ 
40:   Choose step size  $\eta$  satisfying the stochastic Armijo condition:
41:    $T(\theta_{\text{old}} - \eta g) \leq T(\theta_{\text{old}}) - \ell_1 \eta \|g\|^2 + 2 \widehat{\text{Var}}[\epsilon]$ 
42:    $\theta_{\text{new}} \leftarrow \theta_{\text{old}} - \eta g$ 
43:    $\theta_{\text{old}} = \theta_{\text{new}}, i = i + 1$ 
44: end while

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1.1.2 Utility Objective

1.1.3 Simulate ealth dyaics under feedback allocatio strategies

1.1.4 Training the ANN to optimize the expected utility of the underlying wealth process